



THE DATA & AI CHALLENGE · REDROB × INDIA RUNS

Parakh

/pə-rəkh/ — to assay; to test whether gold is genuine.

A candidate ranker that **verifies** each profile before it trusts it — telling genuine engineers from convincing fakes.

Rayyan Ahmed Shaikh · Team Parakh



Great candidates are missed — not absent, just unseen.

Recruiters scan hundreds of profiles and still miss the right person — because **keyword filters can't see fit.**

They reward whoever lists the most buzzwords, and overlook the engineer who **built the exact system** but described it in plain words.

In a pool of 100,000, the real fits are a needle in a haystack — and half the score rides on the top ten.

100,000

profiles in the pool

~few hundred

genuinely fit the role

Top 10

carry 50% of the score

The dataset is engineered to punish keyword search

● Keyword stuffers

Lists every AI skill; the actual job is Marketing or Ops. Counting keywords ranks the fake at the top.

● Plain-language gems

7,334 people describe real retrieval / ranking work under a non-ML title. Title filters miss them entirely.

● Behavioural twins

Near-identical on paper; they differ only in engagement signals — who actually responds and shows up.

● Honeypots (~80)

Impossible profiles (expert skills, 0 months' use). Rank >10% of them in the top 100 and you're disqualified.

Assay, don't match.

Do all the heavy AI thinking offline; bake it into artifacts a fast, offline ranker reads.

Pre-compute

offline · LLM + GPU allowed

Embed + features



Qwen-235B teacher
grades fit 0–5



DeepSeek
2nd judge

Ranking step

offline · CPU · no network · < 5 min

Integrity Gate



Score all
100,000



Top 100
+ reasons

Spec §3 compliance: no LLM, GPU or network during ranking — every model call lives in pre-compute and ships as a cached artifact.

Reject the impossible and the fake — first



Honeypot check

'Expert' in a skill used 0 months · tenure longer than the whole career · contradictory dates. ~84 hits ≈ the ~80 planted honeypots.



Stuffer check

Many AI skills, but the career narrative shows none of the work, and the title is wrong. Keywords without evidence don't count.

VERDICT AT RUNTIME

Flagged profiles are forced to the bottom
— they can never reach the top 100.

0

honeypots in our top 100



A senior recruiter's judgment — that runs offline

01

Grade offline

Qwen3-235B scores the 3,000 most plausible candidates on the JD rubric — fit tier 0–5 with a written reason.

02

Bake in

Those labels are saved as a cached artifact in the repo — a precomputed feature, not a live call.

03

Read at runtime

The offline ranker reads the labels; no LLM, GPU or network is touched while ranking.

Why it matters: an LLM call per candidate can't scale or meet the 5-minute CPU budget. Distillation keeps the intelligence, drops the cost.

Two independent judges — the top is where they agree

Qwen-235B

primary teacher



DeepSeek-V4

second judge (harsher)



Average the tiers

Only candidates **both** rate tier-5
rise to the very top.

The check that pays off

DeepSeek rated only **40 of our 100** as
tier-5. Requiring agreement filters out
one model's overconfidence.

An echo of my ***Quorum*** project: five models debating — trust rises where independent judges converge.



Beyond the résumé

Availability signals

- Recruiter response rate — will they reply?
- Last active — a 6-month ghost isn't hireable
- Open-to-work + notice period
- Interview completion, saved-by-recruiters

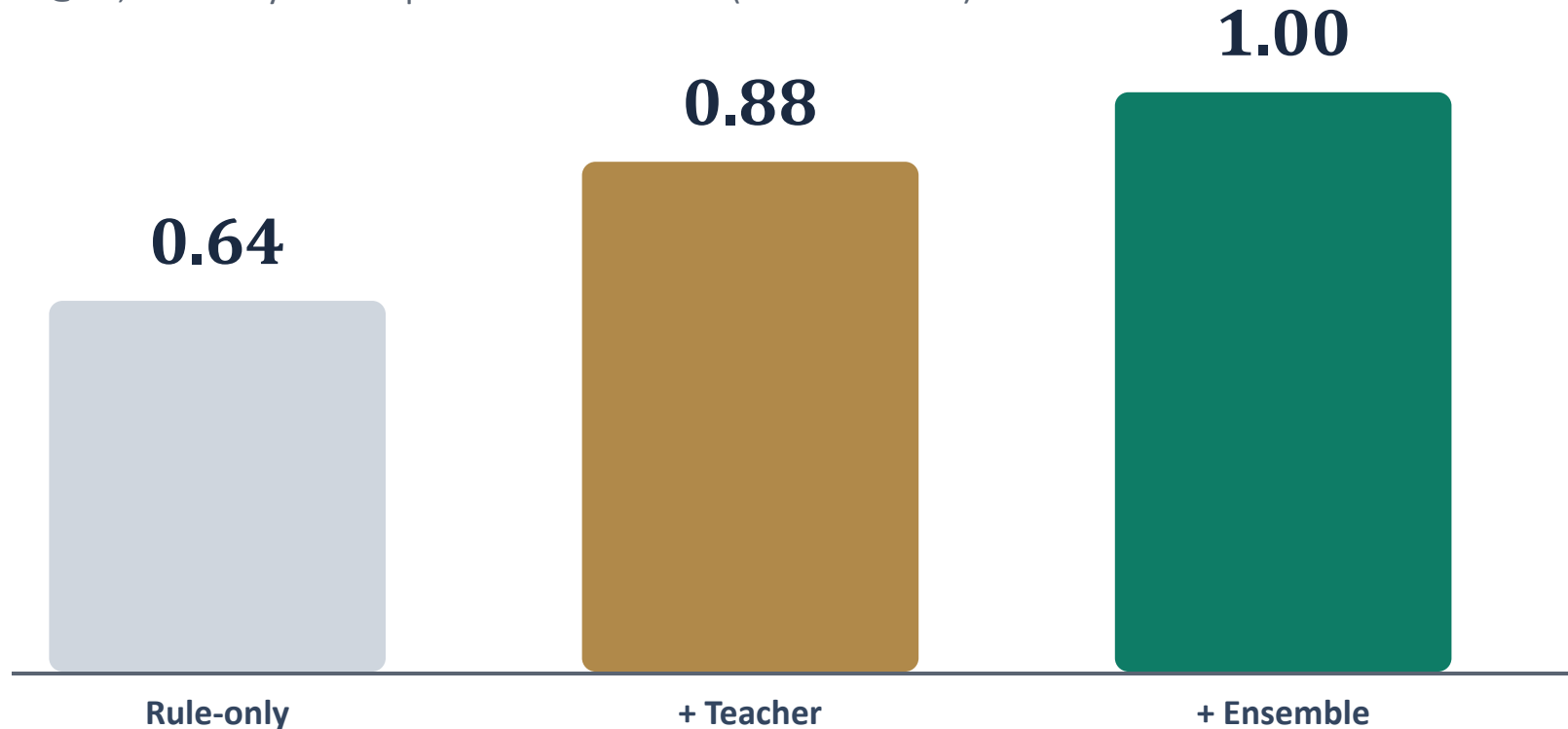
JD hard-negatives

- Whole career at IT-services / consulting
- Pure research, no production deployment
- Based abroad and won't relocate
- 'Architect' who hasn't coded in 18 months

These sharpen precision at the very top, where the score is decided.

Every layer earned its place

NDCG@10, scored by an independent third model (Llama-3.3-70B)



10 / 10

top-10 are tier-5 by both judges

0.98

NDCG@50 (ensemble)

Honest read: *three different models agree the top-10 is excellent — a strong proxy, not the organisers' hidden ground truth.*

Built to survive every stage of judging

- **Reproducible & offline**
The ranking step runs CPU-only, no network, under 5 minutes — it passes Stage-3 reproduction unchanged.
- **It reads profiles**
Zero honeypots and zero stuffers in the top 100 — the system understands, it doesn't pattern-match.
- **Measured, not guessed**
A three-model eval harness gated every upgrade; nothing shipped without a proven gain.
- **Defensible**
Every layer is explainable in plain words — ready to walk through in the finalist interview.



Parakh

Assaying talent — genuine fits, surfaced and trusted.

```
python rank.py --candidates candidates.jsonl
```

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